



CSE260: Digital Logic Design
Summer 2025
Quiz - 03
Total Mark: 15
Duration: 20 Minutes

A

Name:	ID:	Section:
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- 1) Let's assume you have two 4-bit numbers, **A** and **B**. **Design** a circuit using parallel adder-cum-subtractor(s) that will perform the following operations in the order mentioned.
- If A is divisible by 9 and B is less than 3, then perform **B-A**
 - Perform **A+B** otherwise

You may use n-bit parallel adder-cum-subtractor(s), where n is a positive integer. Unused input connection(s) should be connected to ground. Mention MSB and LSB accurately. [10 Mark]

- 2) **Implement** the following function using 4:1 multiplexer only: [5 Mark]

$$F(A,B,C,D) = \Sigma(0,2,5,8,11,15)$$